

11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

1.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer’s unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05’s adjustment tools are powerless here.

1.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs three things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; $F < 10$ is a “weak instrument” danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).

1.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the “compliers,”* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

On real data. IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement** / **intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument.

1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved **engagement** drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized **encouragement** (the instrument) nudges exposure without touching sales directly.

The data-generating model — exactly what `dgp.iv_ad_exposure` implements (defaults & seed in `src/cmp/dgp.py`). $n = 3000$; **unobserved** intent $U = \text{engagement} \sim \mathcal{N}(0, 1)$; randomized encouragement $Z \sim \text{Bernoulli}(\frac{1}{2})$; $\sigma(z) = 1/(1 + e^{-z})$:

$$\begin{aligned} X &= \text{exposure} \sim \text{Bernoulli}(\sigma(0.3 + 1.1 Z + 0.8 U)) && \text{(first stage)} \\ Y &= \text{sales} = 50 + 15 X + 12 U + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 6^2) && \text{(true effect: the 15).} \end{aligned}$$

U appears in **both** equations — $0.8 U$ pushes exposure and $12 U$ pushes sales — that is the self-selection, and since U is unobserved no regression adjustment can fix it (OLS is biased upward). The three IV conditions are visible by construction: **relevance** ($1.1 \neq 0$ — testable, the first-stage F below), **exclusion** (Z appears nowhere in the sales equation — untestable in real data, stress-tested in step 6), and **monotonicity** ($1.1 Z$ only ever *raises* the exposure probability — no defiers).

TRUE exposure->sales effect €15 · first-stage: exposure 56%->77% (shift +21 pp), F = 156

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ($\text{Cov}(\text{exposure}, \varepsilon) \neq 0$), so OLS is biased. A valid instrument Z (encouragement) needs three things:

- **Relevance** — Z moves exposure. *Testable* (the first-stage F above; $F < 10$ = weak).

- **Exclusion** — Z affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** — Z independent of the confounder. Holds by randomization.

What IV identifies — LATE (Imbens–Angrist): under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

Modeling note. Because `ad_exposure` is binary (0/1), this joint MvNormal models the exposure first stage as a **linear-probability (Gaussian)** equation, so the two-equation model is an approximation — the exposure equation’s posterior *width* shouldn’t be read at face value; the reliable read-outs are its point estimate and the endogeneity ρ .

1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.
- **Reduced form** — regress sales on the *instrument* (encouragement $\rightarrow$ sales). This captures only the effect that flows through the clean, randomized channel.
- **First stage** — regress exposure on the instrument (encouragement $\rightarrow$ exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form $\div$ first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity ρ rather than assuming it away).

The fitted models, in symbols. The three frequentist point estimates:

$$\text{OLS: } Y = a_0 + a_1 X; \quad \text{first stage: } X = b_0 + \pi Z; \quad \text{reduced form: } Y = c_0 + \delta Z; \quad \hat{\beta}_{\text{Wald}} = \frac{\delta}{\pi};$$

and the Bayesian IV fits the first stage and the outcome equation **jointly**, with correlated errors:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2 \left(\begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix}, \Sigma \right), \quad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where the off-diagonal ρ is the endogeneity: the correlation between the two equations’ errors that the hidden U induces and that OLS ignores. Priors: weakly-informative $\mathcal{N}(0, 50^2)$ on the regression coefficients (the code comment explains why the library default is too tight here) and an LKJ prior on Σ .

OLS (biased) €23.7

Reduced form (Z $\rightarrow$ Y) €3.5 $\div$ first stage (Z $\rightarrow$ D) 0.21 = Wald/IV ratio €16.5

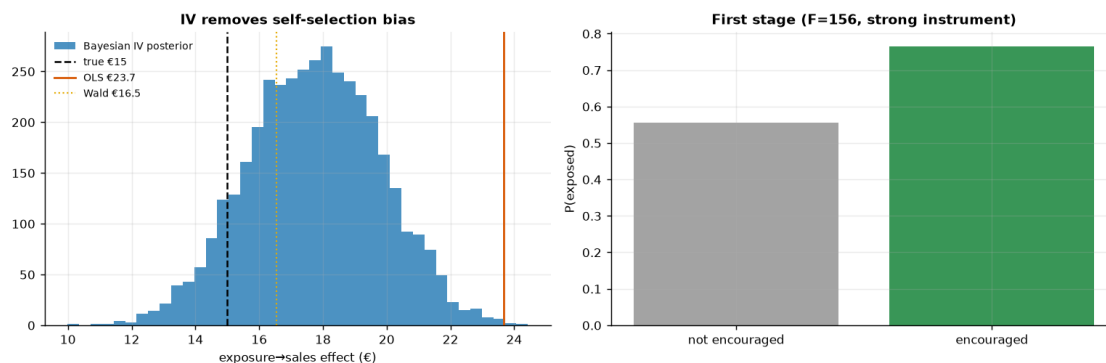
Bayesian IV €17.7 (true €15) 90% credible interval [€14.3, €21.1]
 IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0
 Estimated endogeneity $\rho(\text{exposure err, sales err}) = +0.22$ [90% +0.10, +0.33] -
 nonzero ρ is why OLS (€23.7) sits above IV (€17.7).

1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior’s 90% interval **covering the truth**. The gap between OLS and IV is the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy’s default 2SLS-centred priors would produce.)

One honest note on LATE (and who the compliers are). IV recovers the effect for *compliers* — the customers whose exposure the encouragement actually moves — not the whole population. Under monotonicity that group is exactly the **first-stage shift: +21 pp (56% -> 77%)**, the estimated complier share. In *this* simulation every customer’s true effect is €15 (the effect is homogeneous), so **LATE = ATE** and grading the IV against €15 is a fair test. On real data, where the effect varies across customers, IV would recover the complier-weighted LATE, which generally **!= the ATE** — and that is the right quantity for a budget call, since it answers “*what does an extra exposure do for the customers we can actually move?*”

self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)



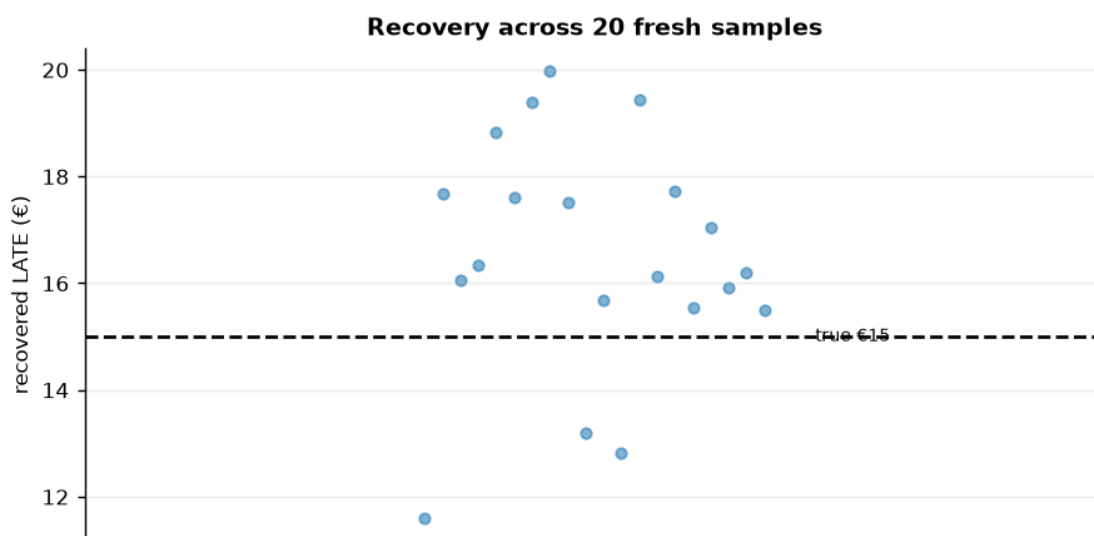
How to read this. *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a small upward finite-sample bias (~€2.7 above on this seed; §5b puts the average bias at +€1.5), but its posterior — now honestly wide, thanks to weakly-informative priors — comfortably **covers €15**; the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of “effect” that were never causal. If you budgeted on OLS you’d over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** ($F \gg 10$). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

1.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each; their sampler chatter is silenced below, and we read only the aggregate bias/coverage tallies) and check both.

IV across 20 seeds: mean €16.5 (true €15) bias +1.5 sd €2.1 · 90% interval covers truth in 16/20 seeds.

The committed seed (37) lands high (€18); across seeds the mean is €16.5 - a small upward finite-sample bias of +€1.5 (~10%) toward OLS that shrinks with n . The 90% interval covers €15 in 16/20 seeds (80% vs the 90% target - modest under-coverage, consistent with that bias); the budget verdict below never hinges on the last euro, which is why the ranking and the cap-style decision survive it.



1.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

At €10/exposure: causal net €7.7 · $P(\text{pays}) = 1.00 \rightarrow$ BUY exposure (to the complier margin)

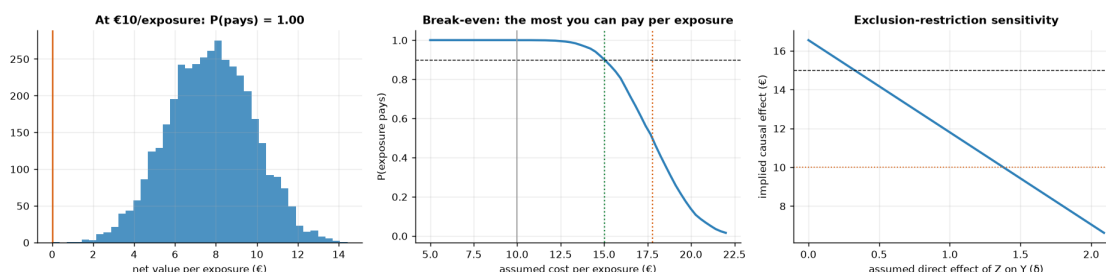
$P(\text{pays})=1.00$ is saturated, not assumed: the whole posterior [€14.3, €21.1] sits above €10.

Break-even sweep: exposure clears the 90%-confidence bar up to €15.0/exposure (the budget cap); at €17.8 it is a coin flip.

So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight. Exclusion stress: the BUY call survives unless the instrument's direct effect exceeds $\Delta \sim 1.4$ (out of a €3.5 reduced form) -

trusting OLS (€24) would overstate the value and overspend.

(the corrected line is a point-estimate bound - it re-solves the Wald ratio at posterior means; carry the IV interval's width alongside it when briefing the cap.)



Read-out. Three numbers make the budget case, one per panel. *Left:* at the current €10/exposure the **causal** net value is ~ €7.7/exposure with the entire posterior above zero — a saturated BUY, valid for the complier margin the instrument identifies. *Middle:* the sweep turns the posterior into a procurement cap — the 90%-confidence bar holds up to ~ **€15/exposure**, the coin-flip point is ~ €17.8, so the €10 plan has ~€5 of genuine headroom. *Right:* the stress test bounds the untestable worry — the BUY survives unless the lottery email itself drives more than $\Delta \sim 1.4$ of sales directly, which against a €3.5 total reduced form would be a large leak (a plain encouragement email with no product content makes that implausible; an email that doubles as an ad does not — design the instrument accordingly). The management contrast to land: budgeting on OLS's €24 would sanction paying up to €24 for something causally worth ~ €15 — the €9 gap, times every exposure bought, is the price of mistaking correlation for incrementality.

1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.